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Design

“There Is An Ongoing Requirement For HIGHER-PERFORMANCE POWER SYSTEM”

All electronic products have power management systems, designing which is a common challenge faced by designers. Recently, Vicor initiated a global survey to understand various aspects of power system design. To understand the survey findings, Rahul Chopra and Sneha Ambastha from EFY spoke to Andy Gales, vice president, Vicor. Excerpts follow:

Q. Do you see Indian companies taking leadership role in power supply design?

A. I think it's already happening and 'Make in India' is the primary example. You have a wide range of engineers in India. Previously, countries like India used to procure power supplies from the US and Europe. However, now they have started designing solutions here themselves.

Q. How important is power supply design in the overall product design cycle?

A. Power supply design is one of the goals but there are other associated tasks too. We found that engineers are commonly multitasking, so often they have more than one power project on the go at a time. Thus, they are not just designing one solution but may be several, for multiple customers. We also found that design engineers are commonly required to develop other parts of the product, sometimes even the software for a new user interface. As a result, very often in these companies, the team building power supplies may not be expert.

In such cases, modular solutions can help them deliver better performance. They don't need to understand details like switching topologies used inside the black box. They can trust the datasheets and rely on the power components to get the performance they require.

Q. How can design engineers achieve higher-performance solutions?

A. There is an ongoing requirement for higher-performance power systems. Achieving high performance from a few resistors, capacitors and FETs can be a struggle without the right expertise. However, if you base the design on power components, which are already optimised for size or efficiency, you can plug these into a circuit board and quickly provide a solution that is close enough to meet the desired product specifications.

Q. What are the common challenges in design engineering?

A. Education. We need to educate engineers to understand the issues faced in using these

components, whether it's meeting EMI/transient input specifications or managing heat. Having the right knowledge is critical for success.

At Vicor, we offer a design review for the common problems that we have seen in the past for power applications. So from the design point of view, we make sure that the customer ends up with a solution that stands the test of time.

Q. What resources are available to help design engineers?

A. Making engineers aware of the best practices and the tools available will help them to get their solution right first time. At Vicor, we have a number of design resources on our website, including our power system designer that helps optimise solutions for size, efficiency or cost. The website also has simulators and whiteboards for estimating the final system performance before you prototype, as well as application notes that cover common issues. We also host seminars and webinars to help engineers understand our technology better. **EFY**